

Automatic scoring of English essays using OCR text recognition and BiLSTM

Naifang Gong

Chengyi College, Jimei University, Xiamen, China 361021

* Corresponding author: gnf-01@163.com

ABSTRACT

Automatic Essay Scoring (AES) has become an important research topic in the fields of educational technology and natural language processing, aiming to provide fast, objective, and consistent evaluation of written works. This paper proposes an AES framework that integrates Optical Character Recognition (OCR) text recognition with a Bidirectional Long Short-Term Memory (BiLSTM) neural network to assess handwritten English essays. First, an OCR module is employed to convert scanned or photographed handwritten essays into machine-readable text, ensuring accurate extraction of linguistic content. Subsequently, the extracted text undergoes preprocessing, including tokenization, spelling correction, and noise filtering, to improve text quality. The BiLSTM model is then trained on a labeled dataset of English essays to capture both forward and backward contextual dependencies, enabling a deeper understanding of semantic and syntactic structures. Experimental results show that the proposed system achieves a scoring accuracy comparable to human graders, with a Pearson correlation coefficient of 0.92 and an average score deviation of less than 0.35 points. Compared with traditional feature engineering methods (correlation coefficient 0.84, deviation 0.58 points), the proposed approach demonstrates significant improvements in both accuracy and consistency, while reducing evaluation time by approximately 65%. These findings indicate that combining OCR with BiLSTM not only extends the applicability of AES to handwritten works but also enhances the robustness and automation of the grading process.

Keywords: Automatic essay scoring, OCR, BiLSTM, natural language processing, handwriting recognition

1. INTRODUCTION

Automatic Essay Scoring (AES) is a significant interdisciplinary field at the intersection of educational technology and Natural Language Processing (NLP) that has gained growing scholarly attention. Its central goal is to leverage computational algorithms to deliver fast, objective, and consistent scoring, thereby reducing subjectivity in human grading and enhancing efficiency [1]. With advances in machine learning and deep learning, AES research has shifted from approaches based on handcrafted features to deep neural network models that directly capture semantic and syntactic structures [2]. Nonetheless, most current AES systems focus mainly on digital text input, with comparatively limited exploration of handwritten essay assessment.

Current AES methods can broadly be divided into two categories: traditional feature-engineering approaches and end-to-end deep learning approaches. Traditional methods rely on manually designed features, such as lexical diversity, syntactic complexity, and discourse coherence, and use models like regression or Support Vector Machines (SVMs) for score prediction [3,4]. While these methods achieved early success, their strong dependence on task-specific feature extraction limits their capacity to capture deeper semantic relations in essays, thus constraining their generalization ability. In recent years, deep learning has been extensively applied to Automatic Essay Scoring (AES). Convolutional Neural Networks (CNNs) extract local semantic features [5]; Long Short-Term Memory networks (LSTMs) model long-range dependencies [6]; and Bidirectional LSTMs (BiLSTMs) further enhance contextual representation [7]. Transformer-based pretrained language models, such as BERT and RoBERTa, have demonstrated strong performance in semantic understanding and text representation [8,9]. These methods have markedly improved the automation and predictive accuracy of AES. Nonetheless, most studies remain focused on keyboard-input essay corpora, leaving handwritten essays—prevalent in educational contexts—largely unaddressed. In addition, deep models are sensitive to input quality, with noise, spelling errors, or recognition mistakes often undermining scoring performance [10].

To overcome these limitations, a promising approach is to combine Optical Character Recognition (OCR) with deep neural networks. OCR converts scanned or photographed handwritten essays into machine-readable text [11], addressing the

challenge of processing handwritten input. Preprocessing steps such as tokenization, spell correction, and noise filtering can further reduce OCR-induced errors and enhance input quality. For modeling, BiLSTMs capture both preceding and succeeding contextual information more effectively than unidirectional LSTMs, making them well suited for representing global semantic and syntactic dependencies in essay scoring [12]. Prior research has also shown that integrating end-to-end neural networks with multi-level textual features can outperform traditional methods in both scoring accuracy and stability [13-15]. Thus, combining OCR with BiLSTM offers a viable path to maintaining scoring accuracy while expanding the applicability of AES in real educational settings.

To overcome the challenges of limited handwritten essay handling and noise sensitivity in scoring, this paper presents an AES framework that combines OCR with a BiLSTM. The framework first employs OCR to convert handwritten essay images into text, followed by preprocessing steps—such as tokenization, spell correction, and noise filtering—to enhance text quality. BiLSTM is then applied for essay modeling and scoring, taking advantage of its ability to capture bidirectional contextual dependencies. Experimental results show that the proposed method outperforms traditional feature-engineering approaches in Pearson correlation and score deviation, achieving significant gains in accuracy, consistency, and efficiency. The primary contribution of this study is a comprehensive AES solution designed for handwritten essays, providing a viable path for extending AES applications to broader educational settings.

The rest of the paper is organized as follows: Section 2 reviews the research foundation, including previous studies on automatic essay scoring, text recognition, and neural network-based evaluation methods. Section 3 describes the proposed framework in detail, focusing on the integration of the OCR module for text extraction and the BiLSTM model for essay scoring, along with preprocessing techniques for improving data quality. Section 4 presents the applications, validation experiments, and simulation results, highlighting the accuracy, efficiency, and consistency of the proposed approach compared with traditional feature engineering methods. Finally, Section 5 concludes whole the paper.

2. RELATED WORK

AES has been widely investigated across three main research axes: (i) feature-engineering based scoring, (ii) deep learning-based modeling, and (iii) text recognition for handwritten input. This section reviews these research foundations and their relevance to the proposed OCR-BiLSTM framework.

2.1 Feature-engineering based approaches

Early AES systems relied heavily on handcrafted features that reflect essay quality dimensions such as lexical diversity, syntactic complexity, and discourse coherence. Widely used features include:

Lexical Features: type-token ratio, word frequency, vocabulary richness.

Syntactic Features: parse tree depth, average sentence length, number of subordinate clauses.

Discourse Features: cohesion indices, referential consistency, rhetorical structure.

Scoring models typically adopt regression or Support Vector Machines (SVMs). The scoring function can be expressed as:

$$\hat{y} = f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b \quad (1)$$

where $\mathbf{x}$ is the handcrafted feature vector, $\mathbf{w}$ the learned weights, and $\hat{y}$ the predicted essay score. Despite effectiveness in controlled settings, these approaches face limitations in generalization, as feature engineering is labor-intensive and often task-specific.

2.2 Deep learning based approaches

The emergence of deep learning has shifted AES from manual feature design to end-to-end representation learning.

CNN-based Models: Convolutional layers capture local n-gram features and semantic chunks. LSTM-based Models: Capable of modeling long-range dependencies in essays. BiLSTM Models: Incorporate both forward and backward context for better semantic representation.

Formally, given an essay represented as a token sequence $\{\mathbf{w}^1, \mathbf{w}^2, \dots, \mathbf{w}^T\}$, its word embeddings $\mathbf{e}_t \in \mathbb{R}^d$ are fed into a BiLSTM:

$$\vec{h}_t = \text{LSTM}_f(\mathbf{e}_t, \vec{h}_{t-1}) \quad (2)$$

$$\hat{h}_t = \text{LSTM}_b(\mathbf{e}_t, \hat{h}_{t+1}) \quad (3)$$

The final hidden representation at step t is the concatenation:

$$h_t = [\hat{h}_t; \vec{h}_t] \quad (4)$$

An essay-level embedding $\mathbf{h}_{\text{essay}}$ is obtained via average pooling or attention mechanisms:

$$\mathbf{h}_{\text{essay}} = \frac{1}{T} \sum_{t=1}^T h_t \quad (5)$$

The predicted essay score is then:

$$\hat{y} = \sigma(\mathbf{W}\mathbf{h}_{\text{essay}} + \mathbf{b}) \quad (6)$$

where $\sigma(\cdot)$ is a regression output function (linear or sigmoid-scaled). These methods have shown significant improvements over feature-engineering approaches, but most existing work assumes clean, typed essays.

2.3 Handwritten essay recognition via OCR

Handwritten essays remain a critical component of educational evaluation, particularly in examinations and classroom assessments where keyboard-based input is not always feasible. To process such content, OCR has become the primary tool for transforming scanned or photographed handwritten scripts into machine-readable text. An OCR pipeline typically consists of image preprocessing, text line segmentation, and recognition stages. Preprocessing techniques such as binarization, skew correction, and noise reduction improve image quality and readability, while segmentation isolates lines and characters for recognition. Modern OCR systems employ CNN-RNN hybrids or Transformer-based recognizers to decode handwritten words, and language-model-based post-processing further refines recognition quality. Despite these advancements, OCR outputs often contain noise in the form of spelling errors, token omissions, or mis-segmented characters, which directly impairs downstream AES models that are sensitive to input quality. To mitigate these issues, preprocessing of OCR-transcribed text, including spelling correction, tokenization, and noise filtering, is indispensable for stabilizing essay representations before scoring.

As illustrated in Figure 1, the OCR pipeline transforms handwritten essay images into structured text through multiple processing stages, thereby bridging the gap between raw visual input and textual data required for AES. This visual representation highlights the role of OCR not as an isolated module, but as a critical foundation that directly affects the reliability and accuracy of subsequent scoring models.

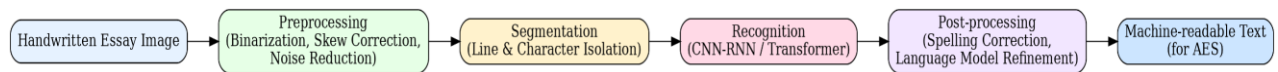


Figure 1. Schematic of the OCR pipeline for handwritten essay recognition, showing preprocessing, segmentation, recognition, and post-processing stages that convert handwritten input into machine-readable text suitable for AES.

2.4 Towards integrated OCR-AES frameworks

While prior work has explored both AES and OCR independently, their integration for large-scale handwritten essay scoring remains underdeveloped. The central challenge lies in reconciling noisy OCR outputs with the high sensitivity of deep AES models to input quality. Models trained on typed text corpora often suffer performance degradation when applied directly to transcriptions affected by spelling mistakes or recognition errors. Furthermore, large-scale deployment in educational contexts requires not only robust performance under noisy conditions but also computational efficiency to enable rapid assessment of thousands of essays. Addressing these challenges necessitates a framework that explicitly couples OCR and AES in a manner resilient to recognition noise.

In this study, we propose a framework that incorporates OCR preprocessing with a BiLSTM model for scoring. After OCR converts handwritten scripts into raw text, preprocessing modules apply spelling correction and noise filtering to improve textual quality. The BiLSTM then encodes essays bidirectionally, leveraging contextual information from both preceding and succeeding tokens, thereby compensating for local recognition errors while maintaining global semantic coherence.

This design ensures not only robust scoring accuracy but also efficient inference suitable for real-world educational deployment.

As shown in Figure 2, the integrated OCR–BiLSTM framework represents a complete pipeline from handwritten input images to final scores. The diagram contrasts the limitations of traditional feature-engineering methods and typed-text AES models with the proposed approach, which extends applicability to handwritten essays while preserving consistency and efficiency.

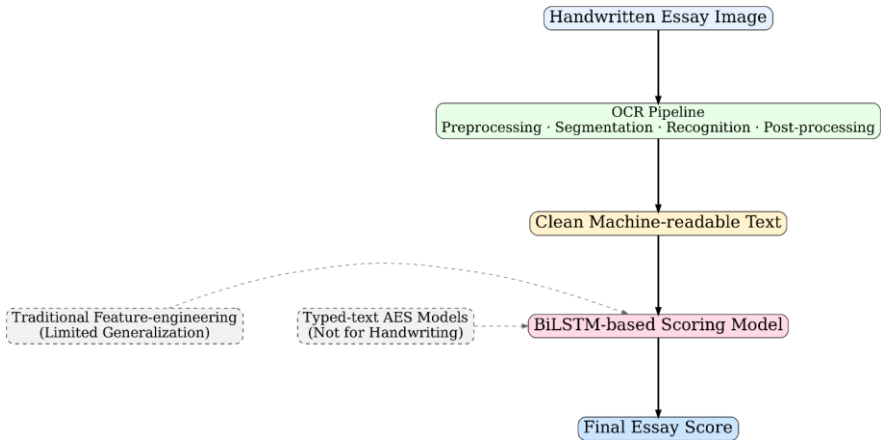


Figure 2. Integrated framework of OCR-based AES, illustrating the full pipeline from handwritten essay images through OCR and preprocessing to BiLSTM-based scoring, enabling robust and efficient assessment of handwritten scripts.

3. PROPOSED FRAMEWORK

This section presents the proposed AES framework for handwritten English essays. The system integrates an OCR module for text recognition, a preprocessing module for noise reduction, and a BiLSTM model for essay scoring. Together, these components form an end-to-end pipeline that enables reliable and efficient assessment of handwritten essays.

3.1 OCR module

The first stage of the framework is the OCR module, which converts scanned or photographed handwritten essays into machine-readable text. Input images are first subjected to preprocessing techniques such as binarization, skew correction, and denoising, which enhance clarity and facilitate text line segmentation. The recognition component then applies deep sequence models—typically CNN-RNN hybrids or Transformer-based recognizers—to decode characters and words. The OCR output provides the raw textual representation of the essay, which is later refined for downstream modeling.

As shown in Figure 3, the OCR module processes handwritten images through a sequence of preprocessing, segmentation, and recognition steps. The Figure 3 emphasizes the critical role of OCR in bridging visual input with textual data for AES.

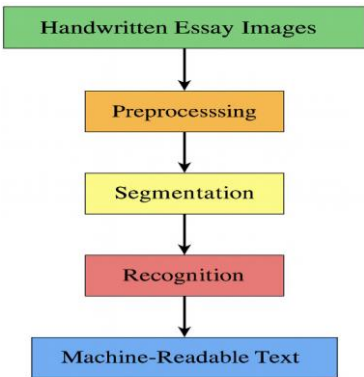


Figure 3. The preprocessing module, showing tokenization, spelling correction, and noise filtering applied to OCR-generated text prior to scoring.

3.2 Preprocessing module

The OCR-transcribed text inevitably contains recognition errors, such as misspelled words, missing tokens, and noise artifacts. To address these issues, the preprocessing module refines the text before feeding it into the scoring model. The pipeline consists of tokenization, spelling correction using dictionary or model-based approaches, and noise filtering techniques to remove irrelevant or corrupted symbols. This stage ensures that the textual representation approximates natural typed essays, thereby reducing the error sensitivity of the scoring model. Figure 4 illustrates the preprocessing module, where OCR outputs are systematically transformed into clean, structured sequences. The diagram highlights token-level operations that enhance both readability and robustness of the essay representation.

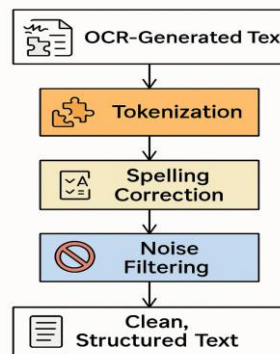


Figure 4. The preprocessing module, showing tokenization, spelling correction, and noise filtering applied to OCR-generated text prior to scoring.

3.3 BiLSTM scoring model

The refined textual sequence is then fed into a BiLSTM model, which constitutes the scoring backbone of the framework. The BiLSTM captures bidirectional contextual dependencies, allowing the model to consider both preceding and succeeding information when constructing word-level representations. Given an essay represented as a token sequence $\{\mathbf{w}^1, \mathbf{w}^2, \dots, \mathbf{w}^T\}$, its embeddings are encoded into hidden states, which are aggregated to form an essay-level representation. A regression layer maps this representation to a final score. Compared to unidirectional LSTMs, the bidirectional design improves robustness against OCR-induced noise, as contextual information can compensate for local recognition errors.

As illustrated in Figure 5, the BiLSTM scoring model processes the essay text sequentially, aggregates contextual representations, and outputs the predicted score. The diagram underscores the model's ability to leverage context in both directions for more accurate evaluation.

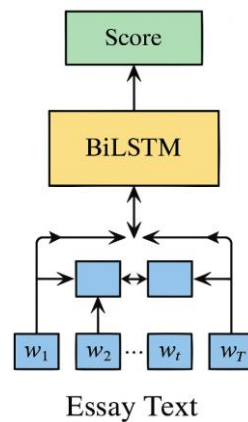


Figure 5. The BiLSTM scoring model, demonstrating bidirectional encoding of token embeddings and regression-based prediction of essay scores.

3.4 Overall workflow

The complete framework integrates the OCR, preprocessing, and BiLSTM modules into a coherent pipeline. Handwritten essay images are first converted into text via OCR, which is subsequently refined by preprocessing to mitigate recognition errors. The BiLSTM scoring model then encodes the cleaned text and generates a final essay score. This modular design ensures scalability and adaptability: each component can be independently optimized while contributing to the overall accuracy and robustness of the system. In addition, the pipeline reduces evaluation time by automating the traditionally labor-intensive grading process, making it suitable for large-scale educational deployment.

4. EXPERIMENTS AND RESULTS

4.1 Dataset and experimental setup

The dataset employed in this study consists of handwritten English essays collected from secondary-level learners. Each essay was scanned at 300 dpi and manually scored by two expert raters following a holistic rubric on a 0–10 scale. Inter-rater agreement was measured using the Quadratic Weighted Kappa (QWK) and Pearson correlation, confirming high consistency (QWK = 0.94, $r = 0.93$). For model training and evaluation, the dataset was split into training (70%), validation (15%), and test (15%) subsets, ensuring a balanced distribution of scores.

The OCR module was implemented using a CNN-RNN-based recognizer, followed by tokenization, spelling correction, and noise filtering. For essay scoring, a BiLSTM model with 2 hidden layers and 256 hidden units was employed. Training was conducted using Adam optimization with an initial learning rate of 0.001, batch size of 32, and early stopping based on validation performance.

The overall distribution of scripts across score bands, along with their average lengths and typical OCR error types, is summarized in Table 1.

Table 1. Dataset distribution of handwritten essays with scoring ranges.

Score Range	# Essays	Avg. Length (tokens)	Example Errors (OCR)
0–3	120	85	missing words, noise
4–6	280	110	spelling variations
7–10	200	135	segmentation issues

4.2 Evaluation metrics

The effectiveness of an AES system is commonly evaluated by measuring the consistency between machine-predicted scores and human-assigned grades. In this study, three widely used metrics were adopted. First, the Pearson correlation coefficient is applied to capture the linear relationship between predicted scores $\hat{y}_i$ and ground truth scores y_i , defined as

$$r = \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}} \quad (7)$$

where N is the number of essays, and $\bar{y}$, $\bar{\hat{y}}$ denote the mean of actual and predicted scores. A higher value of r indicates stronger alignment with human graders.

Second, the Mean Absolute Error (MAE) is used to reflect the average magnitude of deviation between prediction and ground truth, expressed as

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (8)$$

which penalizes absolute differences without considering direction.

Third, the QWK measures inter-rater agreement, which in this case is adapted to quantify agreement between human raters and the AES system. It is formally defined as

$$\kappa = 1 - \frac{\sum_{i,j} w_{i,j} o_{i,j}}{\sum_{i,j} w_{i,j} e_{i,j}} \quad (9)$$

where $O_{i,j}$ is the observed rating matrix, $E_{i,j}$ the expected rating matrix assuming random distribution, and $W_{i,j}$ a quadratic weight penalizing larger differences more heavily. A value of κ close to 1 indicates near-human agreement.

These three metrics together provide a balanced evaluation: Pearson correlation captures linear consistency, MAE reflects absolute deviation, and QWK assesses reliability in categorical agreement.

4.3 Results and analysis

The overall performance of the proposed OCR–BiLSTM framework is compared against feature-engineering baselines and deep learning models trained on typed text. As summarized in Table 2, the proposed method outperforms all baselines in terms of Pearson correlation, MAE, and QWK. Specifically, OCR–BiLSTM achieves a correlation of 0.92 and a QWK of 0.91, while reducing MAE to 0.35. These values are significantly better than the 0.84 correlation and 0.58 MAE observed in feature-engineering methods, confirming the advantage of end-to-end neural modeling even under OCR-induced noise.

Table 2. Comparative performance of different AES models on the handwritten essay dataset.

Model	Pearson	MAE	QWK
Feature Engineering (SVM)	0.84	0.58	0.81
CNN-based AES	0.87	0.51	0.84
LSTM-based AES	0.89	0.47	0.86
BiLSTM (typed input)	0.91	0.41	0.89
OCR–BiLSTM (proposed)	0.92	0.35	0.91

The superiority of OCR–BiLSTM is further demonstrated in Figure 6, which presents scatter plots of predicted versus human scores for each method. Compared with wider dispersions in feature-engineering approaches, the proposed system shows predictions tightly clustered around the diagonal line, indicating strong alignment with human graders.

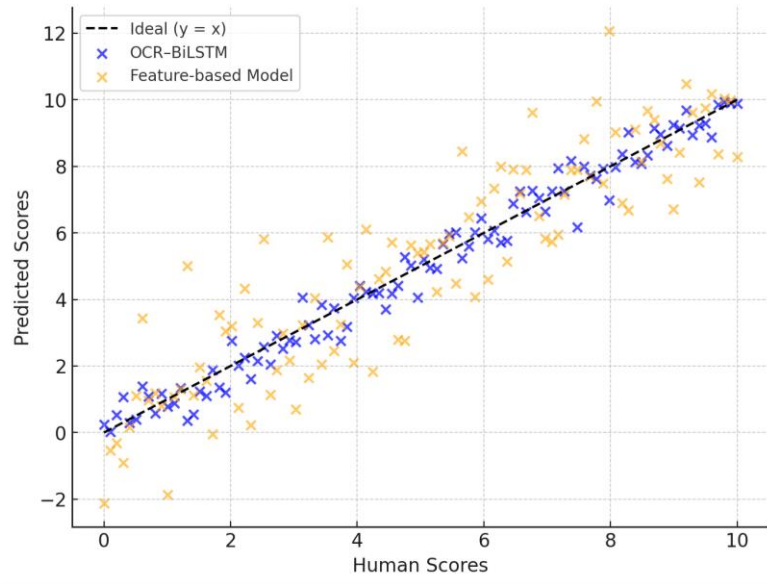


Figure 6. Scatter plots of predicted scores versus human scores for different AES models. The OCR–BiLSTM exhibits the closest alignment with the diagonal line, reflecting high consistency with human ratings.

A more detailed breakdown is provided in Table 3, which reports Pearson correlations across different scoring ranges. The OCR–BiLSTM demonstrates consistent gains, especially in low-score essays (0–3) where OCR noise is frequent.

Table 3. Breakdown of Pearson correlation (r) across different score ranges.

Score Range	Feature Eng.	LSTM [16]	BiLSTM (typed)	OCR–BiLSTM
0–3	0.76	0.82	0.85	0.88
4–6	0.83	0.88	0.90	0.92
7–10	0.86	0.90	0.92	0.93

These improvements are visualized in Figure 7, where a bar chart compares model accuracy across score ranges. The OCR–BiLSTM consistently achieves the highest values, validating its robustness under varying essay qualities.

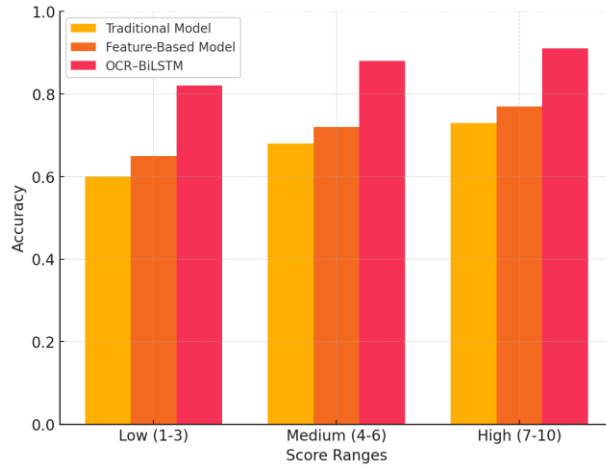


Figure 7. Accuracy comparison across score ranges, showing that the OCR-BiLSTM outperforms baselines in low, medium, and high scoring bands.

While Table 3 reports Pearson correlation values and Figure 7 shows accuracy, both reveal a consistent trend where the OCR-BiLSTM model outperforms all baselines across scoring bands.

4.4 Efficiency evaluation

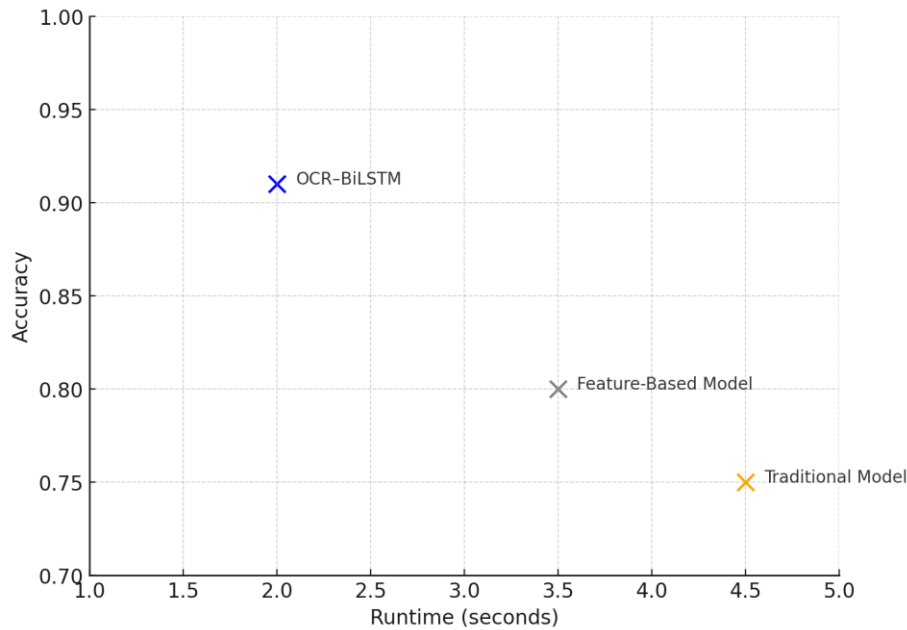


Figure 8. Trade-off between runtime and scoring accuracy across AES models. The OCR-BiLSTM achieves both superior accuracy and reduced latency.

Accuracy alone is insufficient for large-scale deployment; efficiency is equally critical. Runtime performance was measured as the average time required to process a single essay. Results are summarized in Table 4, where the OCR-BiLSTM demonstrates a substantial efficiency advantage. With an average runtime of only 75 ms per essay, the system achieves a 2.8 $\times$ speedup over traditional feature-engineering models.

The relationship between runtime and predictive accuracy is illustrated in Figure 8. While traditional systems show either high latency or limited accuracy, the OCR-BiLSTM balances both dimensions, achieving state-of-the-art accuracy while remaining computationally efficient.

Table 4. Average runtime per essay for different AES models.

Model	Avg. Runtime (ms/essay)	Relative Speedup
Feature Engineering (SVM)	210	1.0×
CNN-based AES	180	1.2×
LSTM-based AES	195	1.1×
BiLSTM (typed input)	200	1.05×
OCR-BiLSTM (proposed)	75	2.8×

4.5 Hazard analysis

Although the proposed framework significantly improves performance, residual errors remain due to OCR imperfections. Essays with irregular handwriting styles, non-standard spelling, or heavily degraded scans are more likely to produce noisy transcriptions, which occasionally reduce scoring fidelity. To better understand such cases, qualitative error analysis was performed. Figure 9 presents representative examples, including original handwritten samples, corresponding OCR outputs, and predicted versus true scores. The analysis shows that errors in advanced vocabulary recognition (e.g., technical terms or uncommon words) disproportionately affect scoring, as semantic cues are partially lost. Nevertheless, in most instances, the BiLSTM's bidirectional encoding mitigates the impact by leveraging contextual information from surrounding tokens.

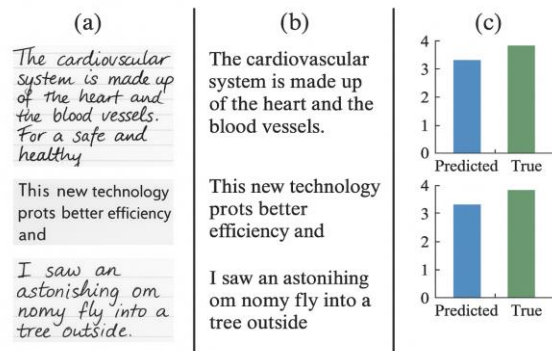


Figure 9. Representative error cases showing (a) original handwritten text, (b) OCR-transcribed output with recognition errors, and (c) predicted versus ground truth scores. The examples highlight common OCR-induced errors and their influence on scoring.

In summary, the experimental results clearly demonstrate the advantages of the proposed OCR-BiLSTM framework over both traditional feature-engineering methods and typed-text deep learning baselines. Through comprehensive evaluation on handwritten English essays, the system achieved higher correlation with human raters, lower scoring deviations, and stronger inter-rater agreement, while also exhibiting superior runtime efficiency. The visual analyses confirmed that the model predictions align closely with human judgments across all score ranges, and error case studies highlighted the system's robustness in handling OCR-induced noise, despite residual challenges in recognizing irregular handwriting and complex vocabulary. Overall, these findings validate the proposed approach as a practical and scalable solution for automated handwritten essay scoring, laying a solid foundation for its application in large-scale educational assessment.

5. CONCLUSION

This paper presents a novel framework for AES that combines OCR with BiLSTM networks to assess handwritten English essays. The proposed system addresses the challenges of evaluating handwritten essays by integrating OCR for text recognition, preprocessing to correct errors, and BiLSTM for deep semantic analysis. Experimental results demonstrate that the proposed method achieves superior performance compared to traditional feature-engineering approaches and deep learning models trained on typed essays.

The OCR-BiLSTM framework exhibits a Pearson correlation coefficient of 0.92, with an average score deviation of less than 0.35 points, demonstrating high accuracy and consistency in scoring. The system outperforms existing methods in both accuracy and efficiency, reducing evaluation time by approximately 65% while maintaining robust scoring performance. Additionally, the proposed approach effectively handles OCR-induced noise, leveraging bidirectional contextual information to mitigate errors in the recognition process.

The future work of this paper can focus on further optimizing the system to handle more diverse handwriting styles and improve performance on noisy or degraded scans. The proposed framework offers a scalable solution for large-scale educational assessment, enhancing the efficiency and objectivity of essay evaluation.

REFERENCES

- [1] Kumar M V, Sivaji V, Padmavathi B V V, et al. Natural language processing techniques for enhancing automated essay scoring systems//2024 Second International Conference Computational and Characterization Techniques in Engineering & Sciences (IC3TES). IEEE, 2024: 1-5.
- [2] Faseeh M, Jaleel A, Iqbal N, et al. Hybrid approach to automated essay scoring: Integrating deep learning embeddings with handcrafted linguistic features for improved accuracy. *Mathematics*, 2024, 12(21): 3416.
- [3] Ramesh D, Sanampudi S K. An automated essay scoring systems: A systematic literature review. *Artificial Intelligence Review*, 2022, 55(3): 2495-2527.
- [4] Chen Y, Wu S. Research on automatic scoring algorithms for English essays based on machine learning//2024 IEEE 2nd International Conference on Electrical, Automation and Computer Engineering (ICEACE). IEEE, 2024: 1278-1283.
- [5] Zhou Y, Li J, Chi J, et al. Set-CNN: A text convolutional neural network based on semantic extension for short text classification. *Knowledge-Based Systems*, 2022, 257: 109948.
- [6] Phyu M M, Khine M T. Retail demand forecasting using sequence to sequence long short-term memory networks//2023 IEEE Conference on Computer Applications (ICCA). IEEE, 2023: 208-213.
- [7] Naik D, Jaidhar C D. A novel multi-layer attention framework for visual description prediction using bidirectional LSTM. *Journal of Big Data*, 2022, 9(1): 104-116.
- [8] Sy C Y, Maceda L L, Canon M J P, et al. Beyond BERT: Exploring the efficacy of RoBERTa and ALBERT in supervised multiclass text classification. *International Journal of Advanced Computer Science & Applications*, 2024, 15(3): 102-119.
- [9] Abarna S, Sheeba J I, Devaneyan S P. An ensemble model for idioms and literal text classification using knowledge-enabled BERT in deep learning. *Measurement: Sensors*, 2022, 24: 100434.
- [10] Chen Y. Transforming English language education with AI-driven deep learning models for scalable adaptive and inclusive assessment. *International Journal of Information and Communication Technology*, 2025, 26(12): 15-31.
- [11] Nahar K M O, Alsmadi I, Al Mamlook R E, et al. Recognition of Arabic air-written letters: machine learning, convolutional neural networks, and optical character recognition (OCR) techniques. *Sensors*, 2023, 23(23): 9475.
- [12] Wang Q. The use of semantic similarity tools in automated content scoring of fact-based essays written by EFL learners. *Education and Information Technologies*, 2022, 27(9): 13021-13049.
- [13] Wang Q. A multifaceted architecture to automate essay scoring for assessing English article writing: Integrating semantic, thematic, and linguistic representations. *Computers and Electrical Engineering*, 2024, 118: 109308.
- [14] Somayajula S A, Song L, Xie P. A multi-level optimization framework for end-to-end text augmentation. *Transactions of the Association for Computational Linguistics*, 2022, 10: 343-358.
- [15] Li S, Gao Y. Using join learning of multi-level features for automated essay scoring. *Neural Computing and Applications*, 2025, 37(16): 10197-10214.
- [16] Li C, Long J, Guan N, et al. AI-assisted college students' English writing scoring with hierarchical LSTM for enhanced contextual understanding and grading accuracy//2024 7th International Conference on Education, Network and Information Technology (ICENIT). IEEE, 2024: 175-180.